

Venkatesh Addala

Orlando, FL, USA · +1-3213169856 · Venkatesh.addala98@gmail.com · [Linkedin.com/in/VenkateshAddala](https://www.linkedin.com/in/VenkateshAddala) · github.com/VenkateshAddala · [VenkateshAddala.online](https://venkateshaddala.online)

Education

University Of Central Florida
Master of Science in Computer Science

August 2024 - May 2026
4/4 CGPA, Orlando, USA

Technical Skills

- **Programming Languages:** Java (Primary), Python, SQL, Scala (Familiar)
- **FullStack:** React, Spring Boot, Microservices, REST APIs, Distributed Systems, Spring Cloud, OAuth2, Concurrency, System Design
- **Data Engineering & Streaming:** Spark, Kafka, Flink, Airflow, dbt, Databricks, Medallion Architecture, Data Observability
- **Databases & Storage:** PostgreSQL, MongoDB, Snowflake, Redis
- **AI/ML:** Applied GenAI (LangChain), Agentic Workflows (MCP), RAG Pipelines, Vector DBs (Pinecone, Milvus), Pydantic, DSPy, Local LLM Integration (Qwen, LLaMa), NLP workflows, Sentence Embeddings, RPA (Robotic Process Automation)
- **Cloud & Containers:** AWS (S3, EC2, IAM, Glue, Redshift), GCP (Dataproc), Azure (ADF Pipelines), Docker, Kubernetes
- **DevOps & Engineering Practices:** Git, Linux, CI/CD (Jenkins/GitHub Actions), Debugging, Performance Optimization

Professional Experience

University of Central Florida
Graduate Research Assistant

August 2025 - May 2026
Orlando, USA

- Architected and implemented an end-to-end **RPA-enabled automation platform** using Python, Flask, and **Playwright MCP** to parse web content, generate student-specific files, automate storage and annotation, reducing manual effort by **90%**, improving **throughput by 10x**, and cutting processing time from **40 minutes to under 3 minutes**.
- Launched a secure student-facing portal integrated with the platform, allowing **200+ students** to access their processed files via OTP-based institutional email authentication, eliminating manual file sharing and improving accessibility.
- Engineered NLP and RAG-based pipelines using LLMs, embeddings, semantic similarity, topic modeling, LangChain, and Pydantic to analyze **100+ student reflections**, retrieve and structure educational data, and reduce manual annotation effort by **~90%+** through intelligent data processing and validation
- **Co-authored accepted ASEE 2026** (NSF Grantees Poster Session) and **AIED 2026** submission on LLM-driven learning analytics
- **Tech Stack:** Python, Flask, Playwright (Chromium), MCP Automation, LLMs (Mistral, Qwen, LLaMA), NLP, Pandas, Matplotlib

Impetus Inc.
Data Engineer

May 2022 - June 2024
India

- Improved processing **performance by ~90% and scalability by ~90%** by converting SAS stored procedures to PySpark on Databricks and deployed them via Azure ADF pipelines.
- Migrated **10+ ETL procedures** from Informatica and Teradata to Snowflake, **improving query performance by ~85%** and **reducing storage costs by ~75%**
- Modernized SSIS workflows by building cloud-based ETL pipelines using Python, PySpark, and Spark SQL on Databricks, **reducing execution time by ~80%**.
- Designed and validated POCs across AWS (Glue, S3, Redshift) and Delta Lake, driving adoption of modern ETL architecture **across 3+ production systems**
- **Tech Stack:** Python, PySpark, SQL, Snowflake, Databricks notebooks, Azure ADF pipelines, Informatica, Teradata, SAS.

Exafluence Inc.
Junior Software Engineer

August 2020 - May 2022
India

- Engineered **EXF-FHIR full-stack application** managing **200M+ healthcare records** with RESTful APIs for CRUD operations, version control, and HL7 FHIR compliance.
- **Built Spark and Kafka ETL modules** for **batch and real-time data ingestion, transforming and loading** data into FHIR-compliant formats
- Redesigned the 1199 Enterprise Data Warehouse by converting complex SQL stored procedures into scalable processing workflows, boosting processing efficiency by 74% and system scalability by 59%
- Owned 2 end-to-end deployments and maintenance of production applications using Linux, Docker and CI/CD pipelines
- **Tech Stack:** Java, Spark, SQL, Hive, Airflow, Spring Boot, React, Kafka, Dataproc, MongoDB, ReactJS, Python, SSIS, Linux, CI/CD

Awards and Certifications

- **Microsoft Certified:** Fabric Data Engineer Associate (2024)
- **Databricks Certified** Developer for Apache Spark (2023)
- **MongoDB Certified** Developer Associate (2021)
- ASEE 2026 (Accepted - NSF Poster Session), AIED 2026 (Under Review)
- **Technical Excellence Award** - Exafluence (2021)
- **Gold Medalist** (Valedictorian - Diploma)